# Joanathan Packia Singh

B.Tech Computer Science & Engineering (Artificial Intelligence)

hello@joanathan.in | joanathan.in | github.com/JoanathanPS | linkedin.com/in/joanathanps

## Summary

Third-year B.Tech CSE (AI) student and Software Development Intern at Zoho. I build and ship full-stack and AI products, won one hackathon and placed top 5 in another.

## Education

**Saveetha Institute of Medical and Technical Sciences (SIMATS)** Chennai, India
*B.Tech Computer Science and Engineering (Artificial Intelligence) | CGPA: 8.8/10* *2024 – 2028 (Expected)*

## Technical Skills

**Languages:** Python, Java, C/C++, JavaScript
**Web:** HTML, CSS, Node.js, REST APIs
**AI/ML:** Generative AI, RAG, Embeddings, OpenCV, NumPy, Pandas, Scikit-learn
**Databases:** PostgreSQL, MySQL, MongoDB, Supabase
**Tools:** Git, GitHub, Linux, VS Code, Vercel, Docker

## Projects

**LoreBook** | *AI Study Workspace* | Next.js, TypeScript, Supabase, Groq, Gemini GitHub | Live

- Built a document pipeline that ingests PDF, DOCX, image, and text files up to 50 MB, chunks and embeds them with Voyage, and serves chat, summaries, flashcards, and quizzes through Gemini.
- Designed around free-tier API and database limits, and improved retrieval quality by moving to a better embedding model.

**ClinDex** | *AI Prescription Review Platform* | Next.js, TypeScript, Supabase, Groq, RxNorm GitHub | Live

- Built a prescription review tool that normalizes medications through RxNorm and uses an AI questionnaire to collect missing patient context before surfacing potential drug interactions.
- Fixed the questionnaire repeating questions that had already been answered, so the flow only asks for information that is still missing.

**Battle Royale** | *Java Game Development* | Java, LibGDX, Supabase GitHub | Play

- Programmed advanced AI bot behavior and enhanced an existing Java-based battle royale game by engineering gameplay systems including loot, safe-zone mechanics, minimap functionality, and settings.
- Refactored and extended the existing codebase with new gameplay behavior, balancing changes, naming, and player-facing features while adapting the original game systems to the updated experience.

**FundMeUp** | *Crowdfunding Web App* | Next.js, React, Tailwind CSS, Supabase, Vercel GitHub | Live

- Designed a crowdfunding application for creating, managing, browsing, and supporting fundraising campaigns, implementing authentication, campaign creation and management workflows, frontend interfaces, and backend/database integration for handling campaign and contribution data.

## Experience

**Zoho Corporation** Chennai, India
*Software Development Intern* *Oct 2026 – Present*

- Developed an email-based sales assistant for Site24x7's Quote24x7 workflow, achieving 99.5% accuracy across 200 multimodal test cases.
- Currently developing Site24x7 monitoring plugins, including a PostgreSQL Cluster plugin, to extend infrastructure monitoring capabilities.

## Achievements & Hackathons

**Farmzy | Smart Agriculture Platform** 2025
*Winner – Next Gen Product Buildathon (SQ1 Security) | Top 5 – Hackforge Hackathon (DDGDVC)*

- Led software development as Software Head, building the frontend, backend, and ThingSpeak API integration for real-time soil monitoring and data visualization, with soil-monitoring hardware feeding live analytics.

**NXTGEN'26 – International Software Innovation Hackathon (SRMIST, Logitech)** 2026

- Developed **AlienBuster**, an invasive species early warning platform combining citizen-reported geotagged images with Sentinel-2 satellite NDVI analysis to identify vegetation anomalies and support early ecological threat detection during a 24-hour hackathon.

## Certifications

**NPTEL Certifications:** Statistical Learning for Reliability Analysis, IoT, Social Network Analysis 2025–2026